DESCENT OF IMMERSIONS ALONG LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Opens of the limit. Limits, Lemma lemma-descend-opens (same chapter)

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01IM: sources/stacks-chow/tags.txt, source lines 1983–1983. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Situation 081D. If*

(1) *f is an immersion, and*
(2) *f_0 is locally of finite type,*

then f_i is an immersion for some $i \geq 0$.

Proof. There exists an open $V \subset Y$ such that the morphism f factors as $X \to V \to Y$ and such that $X \to V$ is a closed immersion, see discussion in Schemes, Section 01IM. Since X is quasi-compact, we may and do assume V is a quasi-compact open of Y. By Lemma 01Z4 after increasing 0 we can find a quasi-compact open $V_0 \subset Y_0$ such that V is the inverse image of V_0. Then the inverse image of V_0 in X_0 is a quasi-compact open whose inverse image in X is X. Hence by the same lemma applied to $X = \lim X_i$ we may assume after increasing 0 that we have the factorization $X_0 \to V_0 \to Y_0$. Then for large enough $i \geq 0$ the morphism $X_i \to V_i$ where $V_i = Y_i \times_{Y_0} V_0$ is a closed immersion by Lemma 01ZP and the proof is complete. □

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